

"We need a vision first": Speculating Deaf-Centered Immersive Classrooms

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Abstract

Extended Reality (XR) technologies are increasingly explored for education, raising questions about how immersive classrooms should be meaningfully designed for minority cultural and linguistic groups such as Deaf communities. This poster presents a prequel that asks what orientations should guide the design of XR classrooms when Deaf perspectives lead. We report on a Deaf-led speculative Future Workshop with four Deaf teachers, who bring long-standing expertise in organizing learning around sign language communication and shared visual attention. Rather than treating Deaf practices as constraints to be accommodated, Deaf teachers envisioned immersive classrooms as relational, collaborative, and interdependent spaces. Our analysis highlights three Deaf-centered design orientations: DeafSpace as ongoing coordination of shared access, sign communication as embodied and relational interaction, and XR as a system for deliberate visual orchestration rather than replication of physical classrooms. This work contributes Deaf-centered conceptual grounding for future XR accessibility research and design.

CCS Concepts

• **Human-centered computing** → **Empirical studies in accessibility**; *Accessibility theory, concepts and paradigms*.

Keywords

Deaf Tech, Extended Reality, Accessibility, Speculative Design

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1 Introduction

Extended Reality (XR) technologies, including virtual, augmented, and mixed reality, are increasingly explored for educational use, raising questions about how space, attention, and interaction are organized in immersive classrooms [2]. Prior work has examined accessibility in XR classrooms and proposed techniques to support diverse needs [1, 3, 21, 45]. However, critical work in human-computer interaction (HCI) has shown that research questions and design priorities are not neutral; whose perspectives are centered fundamentally shapes what problems are identified and addressed [6, 24, 40]. In XR classrooms, this raises a central question: *who defines what matters?* While accessibility-focused XR research often adapts existing classroom models for underrepresented users, such approaches risk overlooking the knowledge, practices, and priorities of the communities they seek to support [23, 28, 30].

This work approaches XR classrooms as Deaf¹ Tech². Deaf education is grounded in sign language communication and shared visual attention [49–51]. Deaf teachers have long-standing expertise in organizing classrooms around visibility, coordination, and collective awareness [4]. They also draw on lived experience of learning in classrooms rarely designed for *Deaf ways of being*. We therefore take Deaf teachers' perspectives as the starting point for imagining future immersive classrooms.

Rather than beginning with predefined hypotheses or XR prototypes, we ask: *what orientations should guide the design of XR*



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¹Capital "D" Deaf refers to individuals who identify with Deaf culture and use sign language as their primary language, while "deaf" or "hard of hearing" often describes audiological hearing status without necessarily implying cultural affiliation [8, 46].

²Angelini defines Deaf Tech as technologies conceptualized, designed, and evaluated from Deaf people's lived experiences, cultural values, and epistemologies, centering Deaf people from the outset [5].

classrooms when Deaf perspectives lead? To explore this, we conducted a speculative Future Workshop with Deaf teachers in Deaf-led classrooms, engaging in collective imagining, discussion, and sketching of ideal immersive learning environments unconstrained by existing technologies and hearing-centered assumptions.

This poster serves as a prequel that presents Deaf-centered design orientations for XR classrooms, grounded in Deaf epistemology and pedagogy during early-stage envisioning. By analyzing Deaf teachers' visions, we frame immersive classrooms as interdependent and collaboratively maintained visual spaces, foreground sign communication as embodied and relational interaction, and identify deliberate visual orchestration³ as a central design opportunity. This work does not aim for generalization, but instead surfaces *Deaf-centered design orientations* that can inform future design and empirical work.

2 Related Work

2.1 Deaf Communities, DeafSpace, and Deaf Education

Deaf studies scholarship has long established Deaf communities as cultural and linguistic minorities centered around sign language, shared social norms, and collective practices, rather than solely an audiological category [36, 37]. Concepts such as *Deafhood* [36] and *Deaf Gain* [10] emphasize the value of Deaf ways of being, while *DeafSpace* articulates principles for designing environments that support visual communication, shared sightlines, and spatial awareness [9, 18, 20]. Together, these frameworks foreground how spatial and interaction design shape Deaf communication and participation.

In Deaf education, instruction delivered directly in sign language is central to full access to learning [4, 29]. Deaf pedagogy is organized around shared visual attention and embodied communication, with classroom practices that prioritize sightlines, coordinated turn-taking, and collective awareness [49, 50]. Teachers actively manage spatial arrangements so participants can see one another, pause to allow visual orientation before interaction, and use spatial referencing within the signing space to anchor concepts [29, 36, 37]. Layouts such as horseshoe-shaped seating arrangements preserve mutual visibility, while pointing, gaze, and bodily orientation are integrated directly with instructional materials [20]. These practices are not accommodations layered onto spoken-language instruction but foundational norms in ASL-first environments. In contrast, many educational contexts involving Deaf learners remain structured around spoken-language expectations, where sign language is mediated through interpretation or accessibility services rather than serving as the primary organizing principle [27] (Fig. 1).

2.2 XR Accessibility for Deaf and Hard of Hearing Users

HCI researchers have explored accessibility in XR systems for Deaf and hard of hearing (DHH) users, reflecting the diversity of communication practices across this population. Prior work has proposed

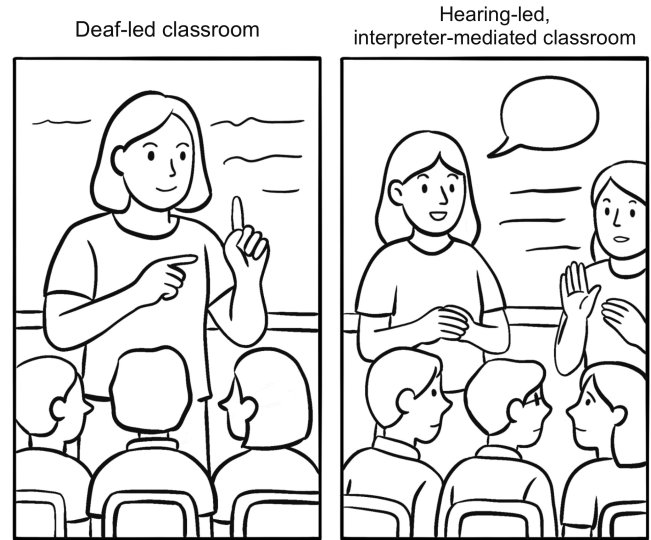


Figure 1: Comparison of classroom dynamics in Deaf-led versus hearing-led and interpreter-mediated classrooms. Aside from instructional materials such as the blackboard, in the Deaf-led classroom (left), all students focus on the Deaf teacher. In the hearing-led and interpreter-mediated classroom (right), students' attention is divided between the hearing teacher, typically positioned at the center, and the interpreter, usually located to the side of the classroom. The illustration was adapted from Huffman et al. [31] and is licensed under CC BY 4.0.

approaches such as captioning [32, 53] and subtitling [1]⁴, interpreter integration [3, 47], sound visualization [26, 33, 38] and modification [15], and visual or haptic representations of sound [17, 34, 43]. Together, these efforts provide important techniques for supporting multimodal perception and situational awareness across diverse DHH users. Much of this work operates within spoken-language classroom structures, introducing accessibility features such as captions or sound representations to enhance inclusion. While these approaches significantly advance XR accessibility, they often retain auditory-centered organizational models rather than reconsidering how classroom interaction itself might be structured from Deaf pedagogical foundations [24, 25]. Limited work has examined how XR classrooms might be designed to support Deaf pedagogy, DeafSpace principles, and sign language interaction as envisioned by Deaf educators themselves.

2.3 Speculative and Participatory Design in HCI

Participatory design has long emphasized the importance of involving underrepresented and underserved groups in the design of technologies that affect their lives [40, 48]. Yet Deaf people are still rarely centered, and even more rarely lead, technology design efforts; even when included, participation is often limited to

³We use "orchestration" to denote the deliberate coordination of visual cues, salience, and timing to support shared attention and interaction in the immersive classroom.

⁴Terminology varies by context, region, and platform. Here, we follow the usage reflected in each paper's title.

late-stage feedback [6]. Such practices risk producing technologies that fail to reflect Deaf epistemologies [7] or that appropriate Deaf cultural practices [23]. Recent work has therefore called for *Deaf Tech*, technologies grounded in Deaf experiences, values, and ways of knowing from the outset [5, 7, 30]. To move beyond the limitations of conventional participatory approaches, researchers have increasingly adopted speculative design, which uses imagined future scenarios to explore values, assumptions, and design possibilities beyond current technological constraints [19, 22, 44]. Speculative design has been used to examine how technology might better serve minority communities, such as through deaf-led participatory workshops grounded in Deaf Tech [7], community-led Afrofuturist toolkits [14], and prototyping activities with neurodiverse dyads [54]. Building on this work, we adopt the Future Workshop [52], a participatory method that supports collective reflection and idea generation, and use it in a speculative manner to explore Deaf teachers' visions of future XR classrooms unconstrained by present technologies.

3 Methods

3.1 Team Positionality

All authors of this paper are Deaf teachers (DT) at a Deaf educational institution, where everyday communication takes place in American Sign Language (ASL) and written English. Our teaching spans disciplines including computer science, HCI, and mathematics, with experience ranging from two to over twenty years in Deaf-centered classrooms. This poster reports on an early-stage internal co-design effort in which four co-authors (DT1–DT4) also served as workshop participants, while the first author facilitated the workshop and led translation and analysis. We intentionally formed an *all-DT author team* and adopted an author-participant model [32, 35] to ground XR classroom design in Deaf teachers' visions and to guide subsequent research. This positionality reflects our methodological goal of treating Deaf teachers not as external informants, but as epistemic leaders shaping the direction of Deaf-centered XR classroom design. While our broader project includes additional workshops with Deaf teachers and students beyond the author team, this poster focuses on these initial sessions, establishing shared problem framings and design priorities for subsequent inquiry.

3.2 The Future Workshop and Qualitative Data Analysis

As a first step in an ongoing effort to center Deaf perspectives in XR classroom design, we conducted a participatory Future Workshop titled *Deaf-led Immersive Classroom*. The goal of these preliminary sessions was to collaboratively identify design priorities for Deaf-centered XR classrooms prior to prototype- or hypothesis-driven work. This early-stage workshop was not intended to represent all Deaf educators, but to establish an initial Deaf-led conceptual grounding. The first author facilitated two workshop sessions with the author team: one with DT1 and DT2, and a second with DT3 and DT4. In each session, Deaf teachers imagined and designed

fictional environments and features for future Deaf-centered immersive classrooms unconstrained by existing technologies or audio-centered assumptions. Designs were shared and discussed collectively, allowing participants to reflect on, critique, and build upon each other's ideas. Each session consisted of four phases: (1) warm-up and context setting, (2) individual envisioning, (3) group co-design and discussion, and (4) collective reflection. All sessions were conducted in ASL and video-recorded with consent.

Workshop discussions and design artifacts were translated from ASL into written English by the first author and analyzed using reflexive thematic analysis [12, 13]. The first author conducted iterative coding of transcripts and sketches, identifying patterns across sessions. Through collaborative discussion with all authors, these codes were refined into three interpretive design orientations that synthesize underlying Deaf pedagogical values. As an all-Deaf author team, we engaged reflexively in interpreting the data.

4 Results and Discussion

4.1 DeafSpace in immersive environment as ongoing visual coordination

Each session began with a discussion of DeafSpace as an environment intentionally designed to support Deaf communication norms. Teachers emphasized that DeafSpace is “*an environment where hearing-based behaviors do not control the classroom*” (DT3), requiring instruction to be organized around visual and spatial communication rather than auditory assumptions. This includes both physical arrangements, such as seating layouts that preserve sightlines, and pedagogical practices that allow time for visual orientation before communication begins. In this sense, DeafSpace was understood not only as a physical configuration, but also as a shared cultural practice in which “*Deaf communication styles are natural and supported*” (DT3).

Building on these understandings, Deaf teachers envisioned immersive classrooms as dynamic and shared visual spaces. They described the need for XR classrooms to support collective awareness of visibility and spatial relationships among teachers and students. In these visions, teachers and students occupy a common visual environment and must remain attentive to how individual actions—such as movement, signer changes, or object placement—affect others' ability to see and participate. DT1 and DT2 emphasized that while individual customization is important, it should not undermine shared understanding. They highlighted the need for systems that help maintain alignment across teachers' and students' views, for example by signaling when signer changes or object movements disrupt visual access (Fig. 2(a, b)). Deaf teachers also discussed how immersive environments could support collaborative spatial arrangements that preserve clear sightlines for everyone. Drawing on familiar Deaf classroom practices such as the horseshoe-shaped layout, DT3 imagined XR configurations that keep the teacher visually centered while grouping students within a shared visual field, making it easier to identify who is signing and to shift attention during discussion (Fig. 2(c, d)).

A central theme concerned the distinction between private and shared visual space. Deaf teachers described wanting shared spaces for instruction and discussion that are accessible to everyone in the class, alongside personal areas for activities such as note-taking or

Deaf teacher sketch (workshop artifact)

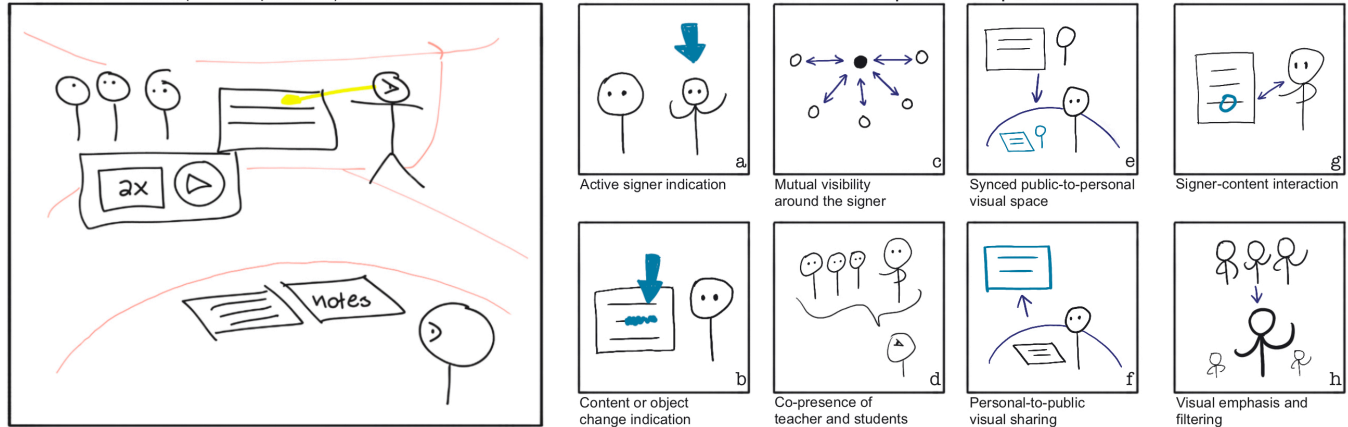


Figure 2: Illustrations of Deaf teachers' visions of Deaf-centered immersive classrooms developed during a speculative Future Workshop. The left shows an original sketch created by DT3. The right presents eight key aspects identified in workshop discussions: (a) Active signer indication, (b) Content or object change indication, (c) Mutual visibility around the signer, (d) Co-presence of teacher and students, (e) Synced public-to-personal visual space, (f) Personal-to-public visual sharing, (g) Signer-content interaction, and (h) Visual emphasis and filtering. Together, these elements reflect core design orientations for Deaf-centered XR classrooms as conceptualized during the workshop.

adjusting accessibility preferences. DT3 further described wanting linked but independently controllable visual surfaces, such as a shared main display paired with personal views in the same visual area, to reduce the need to look back and forth (Fig. 2(e)). DT2 emphasized that while everyone should have access to the shared public space, who has control and when depends on contextual factors such as grade level and course content; for example, a teacher might grant a student permission to share their work in the public space (Fig. 2(f)). Together, these visions highlight DeafSpace in XR as an ongoing process of coordinating shared visual access, balancing private control with collective awareness instead of treating space as a fixed arrangement. DT1 noted, *"in the end, it is still about human-to-human connection. Technology should support that, not replace it."*

Takeaway: Deaf teachers' visions of immersive classrooms highlight DeafSpace as a relational and interdependent practice rather than a fixed spatial layout [11]. In XR classrooms, DeafSpace emerges through ongoing coordination and negotiation between teachers and students as they manage shared and individual visual views, which McDonnell and Findlater describe as "collective communication" [42]. Unlike architectural DeafSpace, immersive environments allow dynamic reconfiguration of shared sightlines and synchronized visual anchoring independent of fixed physical layouts. These findings suggest that Deaf-centered XR design should prioritize mechanisms that maintain alignment, visibility, and shared attention. Instead of requiring Deaf teachers and students to adapt to hearing-centered technologies, XR classrooms should be designed to support Deaf ways of being from the outset, consistent with prior work on Deaf Tech [5, 7].

4.2 Embodied and relational sign communication in XR

Deaf teachers emphasized that immersive classrooms must support sign language as a full-bodied form of communication. They described signing as involving the coordinated use of hands, facial expression, body movement, and spatial alignment with others, all of which contribute to meaning-making during teaching and interaction. They noted that when XR systems isolate or fragment these elements, for example by not providing a view of one's own body, sign communication risks being reduced to something that is technically visible but interactionally diminished. Deaf teachers also stressed that sign communication should remain closely integrated with instructional materials: *"I do not like when the signer is disconnected from the slides or is on a separate screen (DT3)."* They expressed a strong preference for environments in which instructors can physically reference, gesture toward, or interact with slides while signing (Fig. 2(g)). Separating the signer from instructional materials was described as disruptive to teaching, as signing, pointing, and interacting with content are part of a single communicative action.

Deaf teachers closely tied signer representation in XR, such as real people versus avatars and two-dimensional versus three-dimensional representations, to questions of embodiment and trust. DT4 noted, *"I prefer a real person, not an avatar."* While some were open to highly realistic avatars, all expressed concern about whether avatars could accurately convey different signing styles and expressive nuance. They emphasized that representation choices influence how sign communication is perceived and trusted. Deaf teachers also described that observing how students sign and interact provides critical pedagogical feedback, allowing instructors to adjust

explanations and pacing. DT3 further emphasized, in Deaf education, signing styles carry pedagogical and relational information beyond linguistic content alone.

Takeaway: Deaf teachers' visions emphasize that sign communication in XR classrooms must be supported as embodied, relational, and socially situated interaction. These findings suggest that XR design should prioritize full-body visibility, expressive variation, trusted representation, and close integration between signers and instructional materials. In contrast, prior systems have largely approached sign language integration as an add-on feature, rather than supporting the relationship between signers and content [3, 47]. When XR classrooms are designed to support interaction and intentionally leverage XR's unique affordances, features such as embodiment and spatial alignment with content can better align with Deaf ways of communicating and coordinating shared attention.

4.3 XR as Deaf-Centered Visual Orchestration

Deaf teachers described XR's value not as its ability to replicate physical classrooms, but as its potential to organize visual information in ways that are difficult or impossible in real-world settings. Rather than viewing XR as a substitute for in-person instruction, they emphasized leveraging its unique affordances to better support visual attention and interaction: *"We should not try to use it to replicate in-person interaction. Instead, we should focus on what VR does well (DT4)."* Deaf teachers highlighted XR's potential to support visual attention management, noting that Deaf people must constantly shift between multiple visual sources in classrooms, which can be cognitively exhausting. DT2 compared this need for control to adjusting audio volume, describing how visual elements could be "turned up"—for example by enlarging or highlighting the active signer, speaker, or content—or "turned down" by reducing unnecessary visual information to regulate focus (Fig. 2(h)). Everyone agreed that XR could enable both teachers and students to manage visual complexity by flexibly adjusting the prominence of signers, slides, or other visual elements based on situational and pedagogical needs.

At the level of shared space, Deaf teachers also discussed how immersive environments could support shared visual spaces that exceed the constraints of physical screens. They imagined environments where participants could orient toward the same signer or shared object regardless of physical position. DT2 noted: *"the shared digital space could be much more flexible than a physical screen."* Such configurations were described as supporting shared focus without requiring participants to occupy specific physical locations (Fig. 2(d)). Deaf teachers further emphasized that visual orchestration should support Deaf social interaction, including group awareness and conversational flow. Features that support dynamic spatial reorganization and highlight shared objects during discussion were described as important for minimizing distraction and maintaining collective engagement.

Takeaway: Deaf teachers' visions frame XR classrooms as systems for visual orchestration rather than replication. They suggest that XR design should prioritize the deliberate coordination of shared visual attention—helping learners know where to look, when to shift focus, and how interactions unfold—rather than

merely reproducing physical classroom. Prior research shows that because most educational technologies are designed around auditory assumptions, Deaf learners often experience visual attention split, leading to increased cognitive load and reduced learning quality [16, 39, 41]. By enabling flexible organization and visualization of people, space, and information, Deaf-centered visual orchestration highlights XR's potential to support attention, interaction, and participation in ways that extend beyond the limits of physical and video-based classrooms.

5 Conclusion

We therefore answer our guiding question - *who defines what matters?* - by advancing a Deaf-centered vision for immersive classrooms grounded in Deaf teachers' expertise, values, and pedagogical practices. We treat Deaf teachers as epistemic leaders and Deaf practices as the starting point for design. Drawing on this Deaf-led speculative Future Workshop, we surface Deaf-centered design orientations for immersive classrooms guided by Deaf teachers' visions: DeafSpace in XR as relational, collaborative, and interdependent; sign communication as relational interaction; and XR as deliberate visual orchestration of shared attention. We hope this work informs future XR accessibility research and design that **builds with Deaf communities from the outset**.

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A The Future Workshop Protocol: Deaf-led immersive classroom

A.1 Overall Aim

To explore how Deaf teachers envision and arrange immersive (XR) Deaf-led classrooms, highlighting Deaf ways of teaching, visual arrangement, and communication flow in speculative immersive environments.

A.2 Before the Session

Please prepare blank paper (without lines or grids) and a few working pens or markers for sketching. If possible, bring colored pens, as they can help express different ideas or spatial elements. If you prefer to use a digital drawing tool (e.g., Canva, Figma, Apps on iPad), feel free to do so.

Note: The sketches will be collected as part of the research data (metadata). Please ensure your materials are ready and functional before the session begins.

A.3 Session Plan (1 hour)

0–10 min | Warm-Up & Context Setting

Goal: Build comfort, align understanding, and introduce speculative thinking.

Activities:

- Begin recording (after confirming consent).
- Quick introductions: name, teaching background (years and subject area), and what DeafSpace means to them.
- Quick discuss: What are current challenges or frustrations with classroom visual flow or attention in Deaf-led remote classrooms?
- Facilitator introduction to speculative design. “We’re not solving a problem today. We’re imagining possible futures.”
- Show a short example (*CSD Vision Film: The Future*⁵, 3:06–3:22) featuring a remote class with floating visual elements. Emphasize that there are no wrong ideas. We focus on imagination, creativity, and Deaf Teacher-centered perspectives.

10–30 min | Individual Reflection: Envisioning the Future Classroom

Goal: Generate diverse ideas rooted in lived experience.

Activities:

- Invite participants to visualize their ideal immersive classroom.
- Provide simple symbols of common classroom elements to help with envisioning (Fig. 3).
- Participants sketch, sign, or describe their classrooms using paper or digital whiteboards.
- Encourage notes or explanations in ASL or written English, whichever feels natural.

Guiding questions:

- How would you arrange the space for visual clarity and flow?
- Where would the teacher, students, and materials appear?
- What tools or features (e.g. what kind of customizations) would support Deaf ways of teaching and learning in immersive space?

30–45 min | Group Co-Design & Discussion

Goal: Build a shared speculative vision and uncover key values.

Activities:

- Each participant presents their vision (2–3 minutes each).
- Facilitator leads group discussion:
 - What common ideas or challenges appear?
 - How do Deaf pedagogical values shape these imagined spaces?

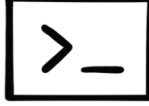
Teacher / Signer	
Slide screen	 or 
Code or content screen	 or 
Wildcard (please add text to describe it)	

Figure 3: Template illustrating the visual elements provided to Deaf teachers for use in speculative designs of immersive classrooms.

- Deaf pedagogical values: Shared sign language, visual access, embodied knowing, community connection, identity affirmation, and respect for visual rhythm.
- Distinguish between the Deaf experience (as teacher or student) and the content, focusing on how the content is delivered visually instead of aurally.
- How might communication, attention, or power shift in immersive classrooms versus traditional face-to-face / remote classrooms?
- Imagine the technology without any limitations.
- As a group, sketch a composite classroom concept combining everyone’s ideas. Use paper or iPad.
- The facilitator captures this visually on a shared Canva whiteboard.

45–55 min | Reflection & Thematic Prompts

Goal: Surface reasoning, emotion, and ethical considerations.

Prompts:

- How does this future classroom *feel* compared to classrooms today?
- What new possibilities or challenges emerge?
- What cultural or ethical values should future immersive technology respect?

⁵<https://www.youtube.com/watch?v=nocgYp5pMG0>